

Predicting song popularity using a Spotify & Youtube music dataset

DSA 406 - Final Project Highlights

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Background

- Several statistics for the Top 10 songs of various spotify artists and their youtube video
- It includes 26 variables for each of the songs collected from spotify and youtube
- Sourced from Kaggle.

Here are the columns in the data set:

[1]	"X"	"Artist"	"Url_spotify"	"Track"
[5]	"Album"	"Album_type"	"Uri"	"Danceability"
[9]	"Energy"	"Key"	"Loudness"	"Speechiness"
[13]	"Acousticness"	"Instrumentalness"	"Liveness"	"Valence"
[17]	"Tempo"	"Duration_ms"	"Url_youtube"	"Title"
[21]	"Channel"	"Views"	"Likes"	"Comments"
[25]	"Description"	"Licensed"	"official_video"	"Stream"

Research Question of Interest

Question of Interest:

Can we predict the popularity of a song based on its audio features (e.g., danceability, energy, valence)?

Hypotheses:

① Audio Features:

- Songs with higher **valence** (positive/happy tone) will be more popular.
- Songs with higher **danceability** and **energy** will have more streams and views.
- Songs with a moderate **tempo** (neither too fast nor too slow) will be more popular.

② YouTube Engagement:

- Songs with **official videos** will have significantly more views and streams.
- Higher **likes** and **comments** on YouTube will correlate with higher Spotify streams.

③ Other Features:

- **Singles** will have more streams and views compared to songs released as part of an album.
- **Licensed content** will perform better due to higher visibility and promotion.

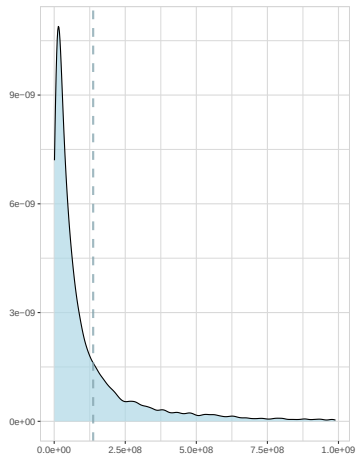
Variables of Interest

Variable	Description
Danceability	A measure of how suitable the track is for dancing, based on tempo, rhythm stability, beat strength, and regularity. Ranges from 0.0 (least danceable) to 1.0 (most danceable).
Energy	A measure of the track's intensity and activity, ranging from 0.0 to 1.0. High-energy tracks feel fast, loud, and noisy, while low-energy tracks are calmer and softer.
Key	The key of the track, represented as an integer using standard Pitch Class notation (e.g., 0 = C, 1 = C / D , 2 = D). A value of -1 indicates no key was detected.
Loudness	The average loudness of the track in decibels (dB), typically ranging from -60 to 0 dB. Higher values indicate louder tracks.
Speechiness	Detects the presence of spoken words in the track. Values above 0.66 indicate mostly spoken content, while values below 0.33 likely represent music or non-speech content.
Acousticness	A confidence measure (0.0 to 1.0) of whether the track is acoustic. A value of 1.0 indicates high confidence the track is acoustic.
Instrumentalness	Predicts whether the track contains no vocals. Values closer to 1.0 indicate a higher likelihood of no vocal content.
Liveness	Detects the presence of an audience in the recording. Values above 0.8 suggest a strong likelihood the track was performed live.
Valence	A measure of the track's musical positiveness, ranging from 0.0 (negative emotions like sadness or anger) to 1.0 (positive emotions like happiness or euphoria).
Tempo	The estimated tempo of the track in beats per minute (BPM).
Duration_ms	The duration of the track in milliseconds.
Stream	The number of streams the song has on Spotify.
Views	The number of views the YouTube video has received.
Likes	The number of likes the YouTube video has received.
Comments	The number of comments on the YouTube video.
Description	The description of the YouTube video.
Licensed	Indicates whether the video represents licensed content, meaning it was uploaded by a YouTube content partner and claimed by them.
official_video	A boolean value indicating whether the video is the official video for the song.

Distribution of Key Variables (1)

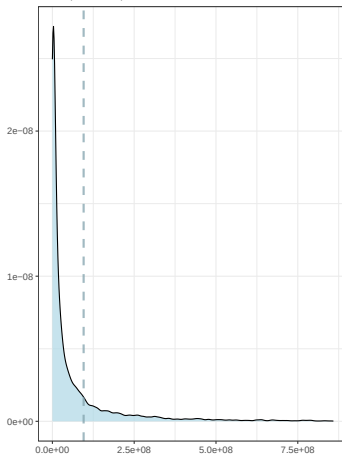
Distribution of Spotify Streams

na = 0, min = 6 574, max = 3 386 520 288



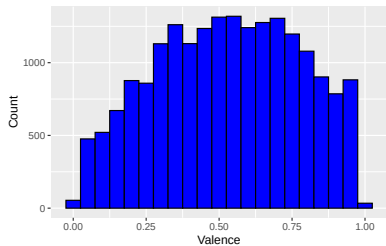
Distribution of Youtube Views

na = 0, min = 26, max = 8 079 649 362

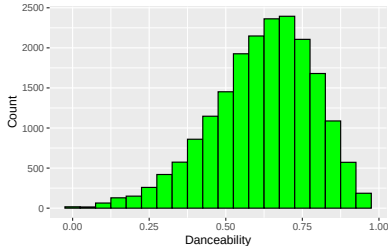


Distribution of Key Variables (2)

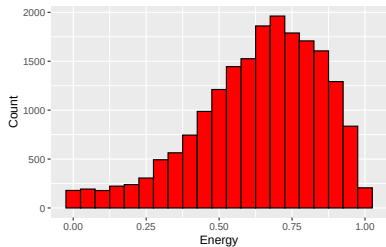
Valence Distribution



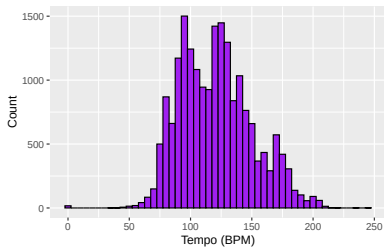
Danceability Distribution



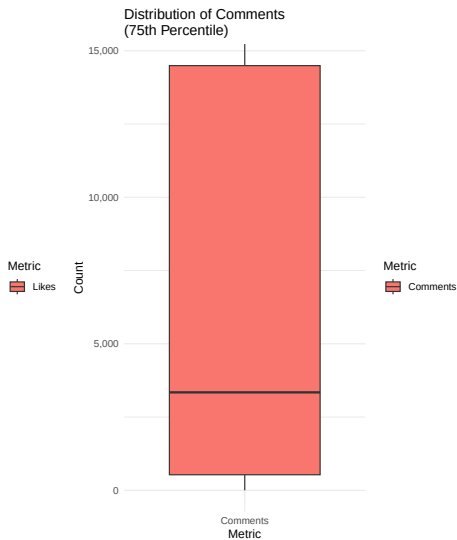
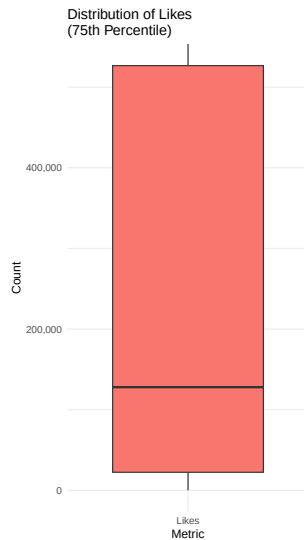
Energy Distribution



Tempo Distribution



Distribution of Key Variables (3)



Data Processing and Data Moves

The `popularity_score` is a value between 0 -100 and is based on the columns Likes, Comments, Stream, and Views .

Developing the Popularity Score

- *Step 1: Normalize the Columns*

$$\text{Normalized Value} = \frac{\text{Value} - \text{Min}}{\text{Max} - \text{Min}}$$

- *Step 2: Assigning Weights and calculate the score*
 - **Views:** 35% weight (since views are a strong indicator of popularity).
 - **Streams:** 38% weight (Spotify streams play a huge role in how much traction an artist has, especially since more people use spotify to listen to music compared to YouTube).
 - **Likes:** 16.5% weight (Engagement is important but not as much as views. Also “likes” don’t determine if a song is viral, songs with a lot of dislikes can also be popular too).
 - **Comments:** 10.5% weight (comments are a weaker indicator of popularity compared to likes)

Data Processing and Data Moves Cont.

Here is what this looks like in R code:

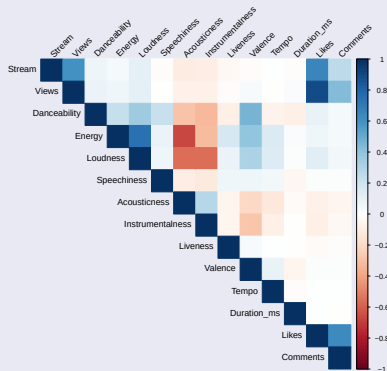
```
# Normalize the columns
normalize <- function(x) {
  (x - min(x, na.rm = TRUE)) / (max(x, na.rm = TRUE) - min(x, na.rm = TRUE))
}

clean_spotify_df <- clean_spotify_df %>%
  mutate(
    norm_views = normalize(Views),
    norm_likes = normalize(Likes),
    norm_streams = normalize(Stream),
    norm_comments = normalize(Comments)
  )

# Calculate popularity score
clean_spotify_df <- clean_spotify_df %>%
  mutate(
    popularity_score = (norm_views * 0.35 + norm_likes * 0.165 + norm_streams
      * 0.38 + norm_comments * 0.105) * 100
  )
```

Investigating Question of Interest

Correlation Matrix



Basic linear regression

Call:

```
lm(formula = popularity_score ~ Danceability + Energy + Loudness +  
    Speechiness + Acousticness + Instrumentalness + Liveness +  
    Valence + Tempo, data = clean_spotify_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.694	-1.914	-1.179	0.182	71.626

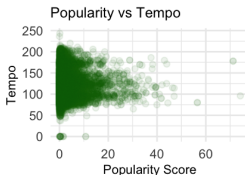
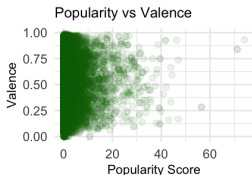
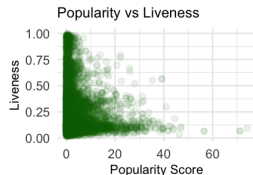
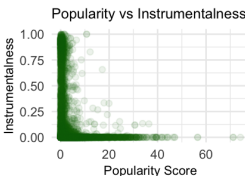
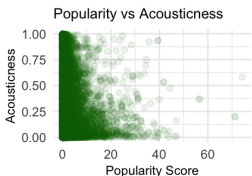
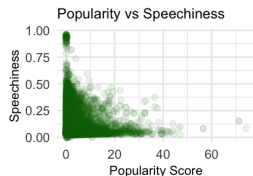
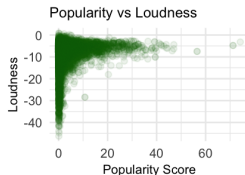
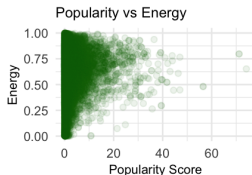
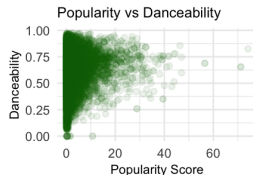
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.6542807	0.3128097	14.879	< 2e-16 ***
Danceability	1.4247543	0.2238982	6.363	2.02e-10 ***
Energy	-1.8499446	0.2542137	-7.277	3.54e-13 ***
Loudness	0.1348613	0.0113779	11.853	< 2e-16 ***
Speechiness	-1.2159925	0.2867987	-4.240	2.25e-05 ***
Acousticness	-1.0997574	0.1415353	-7.770	8.22e-15 ***
Instrumentalness	-0.4908150	0.1905159	-2.576	0.009995 **
Liveness	-0.6692408	0.1828931	-3.659	0.000254 ***
Valence	-0.8509150	0.1473056	-5.777	7.74e-09 ***
Tempo	-0.0008932	0.0010182	-0.877	0.380371

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.091 on 19539 degrees of freedom
Multiple R-squared: 0.03023, Adjusted R-squared: 0.02978
F-statistic: 67.68 on 9 and 19539 DF, p-value: < 2.2e-16

Investigating Question of Interest



Investigating Question of Interest

Hypothesis 2:

Songs with official videos will have significantly more views and streams (and thus higher popularity_score).

Welch Two Sample t-test

```
data: popularity_score by official_video
t = -26.596, df = 14494, p-value < 2.2e-16
alternative hypothesis: true difference in means between group False and group True is not equal to 0
95 percent confidence interval:
-1.423166 -1.227789
sample estimates:
mean in group False mean in group True
1.153143 2.478620
```

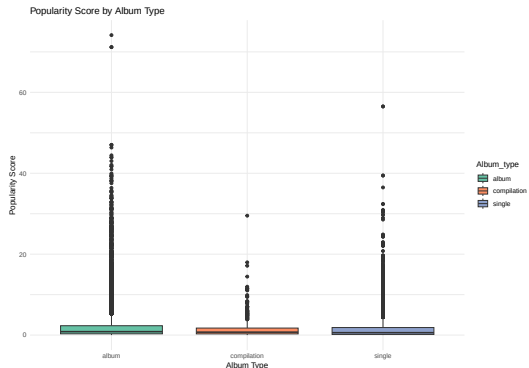
Interpreting the results

Because the P-value is below 0.05, we can reject the null hypothesis (that there's no difference in means) and conclude there's a statistically significant difference in popularity_score between official and non-official videos. The mean in group FALSE and mean in group TRUE show the direction – official videos likely have a much higher mean score.

Investigating Question of Interest

Hypothesis 3:

Singles will have more streams and views compared to songs released as part of an album.



	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Album_type	2	1577	788.4	45.91	<2e-16 ***
Residuals	19546	335664	17.2		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Interpreting the results

The ANOVA results strongly suggest that the album type has a significant impact on the popularity score (streams and views). From the box plot it becomes more clear that that albums garner more “popularity”. The reason for why this might be is because tracks from highly anticipated albums tend to create a sense of “hype” which translates to more views and streams.

Investigating Question of Interest



Investigating Question of Interest

Results of clustering

```
# A tibble: 4 × 12
  cluster Danceability Energy Loudness Speechiness Acousticness Instrumentalness
  <fct>      <dbl>   <dbl>   <dbl>      <dbl>      <dbl>      <dbl>
1 4          0.651 0.748   -5.83     0.0663     0.152     0.0283
2 1          0.723 0.654   -6.89     0.313     0.228     0.00723
3 3          0.541 0.428   -9.90     0.0485     0.561     0.0178
4 2          0.328 0.195  -22.3     0.0460     0.824     0.819

# 5 more variables: Liveness <dbl>, Valence <dbl>, Tempo <dbl>,
# popularity_score <dbl>, count <int>
```

Interpreting the results

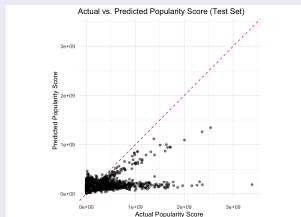
The cluster with the highest popularity score was cluster 4 This grouping involved: - High Energy (0.75) & High Loudness (-5.8 dB) - Low Acousticness (0.15) & Low Instrumentalness (0.03) - Moderate-High Danceability (0.65) & Moderate-High Valence (0.61) - Low Speechiness (0.07) - Moderate Tempo (123 BPM)

Investigating Question of Interest

Creating a Machine Learning model

- 1 Prepare the data
- 2 Split the data
- 3 Train the model
- 4 Make predictions on test set

Results of ML model



Evaluation Metrics on Test Set:

RMSE (Root Mean Squared Error): 220986616

MAE (Mean Absolute Error): 129069674

R-squared: 0.2047

Overall Conclusions

There is no such a thing as a “winning formula” to make the most popular song.

Insights

- 1 Using both the cluster analysis and the multivariable analysis, it was pretty clear that songs with strong danceability, energy, loudness along with a tempo around 130 tend yield higher popularity scores.
- 2 Linear Regression: A linear model predicting popularity_score using audio features and official_video status can explain some variance. Loudness, Energy, Acousticness, and official_video status appear to be significant predictors. However, the model likely violates key assumptions (normality, homoscedasticity) due to the highly skewed nature of popularity, limiting its predictive reliability without further refinement (e.g., transformations, different model types).
- 3 Hypothesis Testing: There's strong evidence that songs with official videos have significantly higher popularity_score.

Do these results make sense?

Yes, The findings largely align with intuition and industry knowledge:

- Higher energy and loudness are common in popular music charts.
- Acoustic or highly instrumental tracks are often more niche than mainstream vocal pop/rock/electronic tracks.
- Official videos are crucial for promotion and engagement.
- The difficulty in perfectly predicting popularity with just audio features makes sense – marketing, artist fame, timing, trends, and subjective taste play huge roles.
- The skewed distribution of popularity (“winner-takes-all” market) is realistic.

Other Considerations

Issue with the data set

The data set used only consisted of top performing songs with various release dates.

How could this project be improved?

To improve this study the data set could also be changed to include all of the music made in a certain year then comparing the streams or views between those songs using the same song component variables to accurately develop a model.

Reflection

This project does not fully answer the originally stated question of being able to accurately predict the popularity of a song, however though this study key factors that make a strong popular song were found.